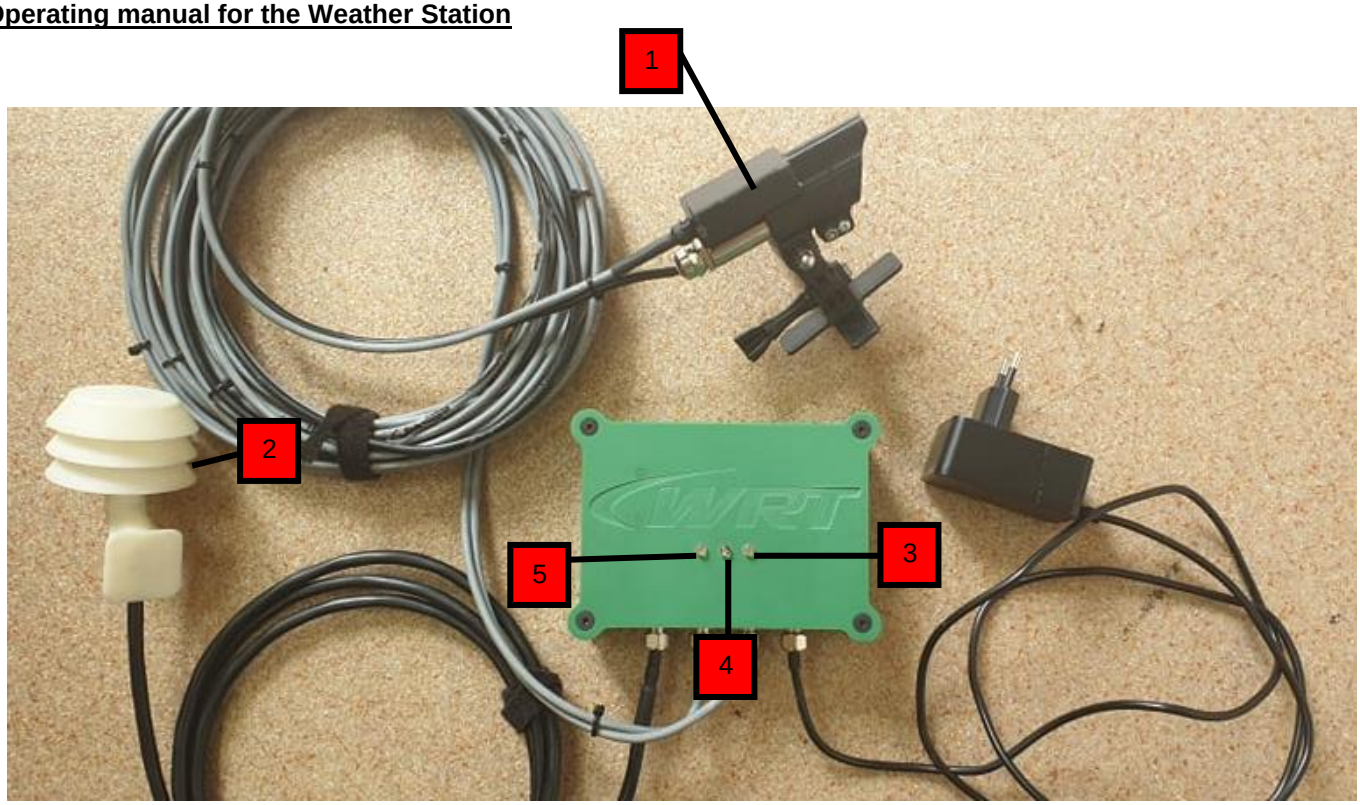
Operating manual for the Weather Station



Nomenclature:

_ 1: The track temperature infrared sensor.

_ 2: The outdoor sensor for temperature, humidity, and atmospheric pressure.

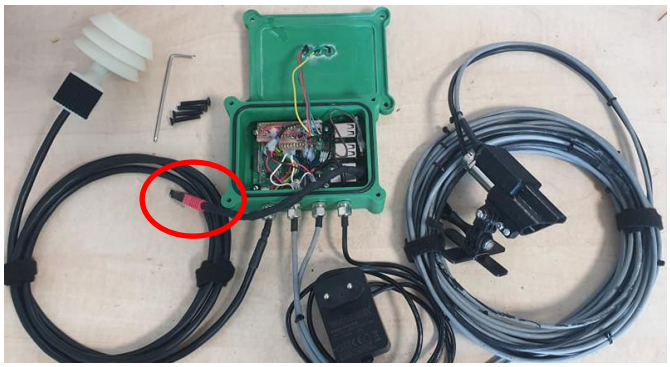
_ 3: Green LED, functional indicator for global system.

_ 4: Orange LED, functional indicator for sensors

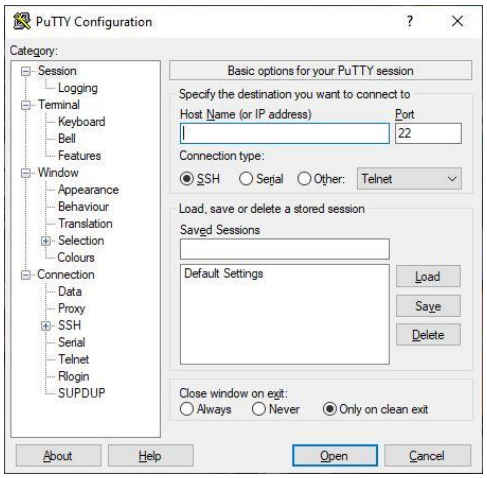
_ 5: Red LED, connection status on network

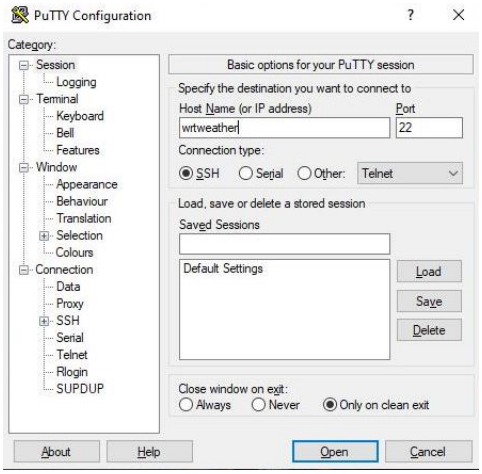
For the first connection on the network

Open the casing of weather station, connected the RJ45 on your computer



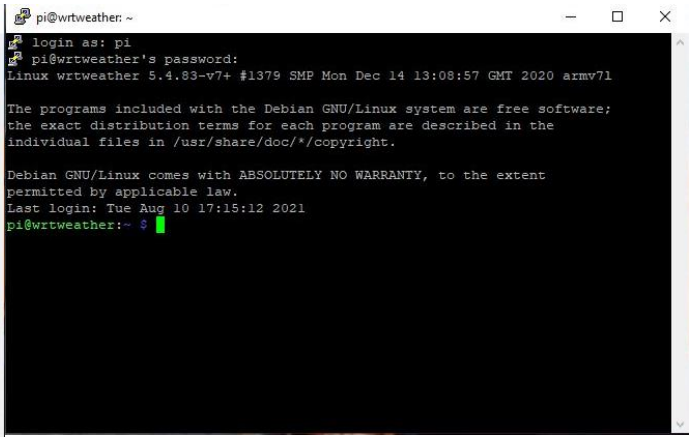
Open Putty software for establishing SSH connection with Raspberry.





When the SSH connected you can enter the login and password:

- Login: pi
- Password: pi



Now you must use the following command line for enable WIFI:

sudo nano /etc/wpa_supplicant/wpa_supplicant.conf

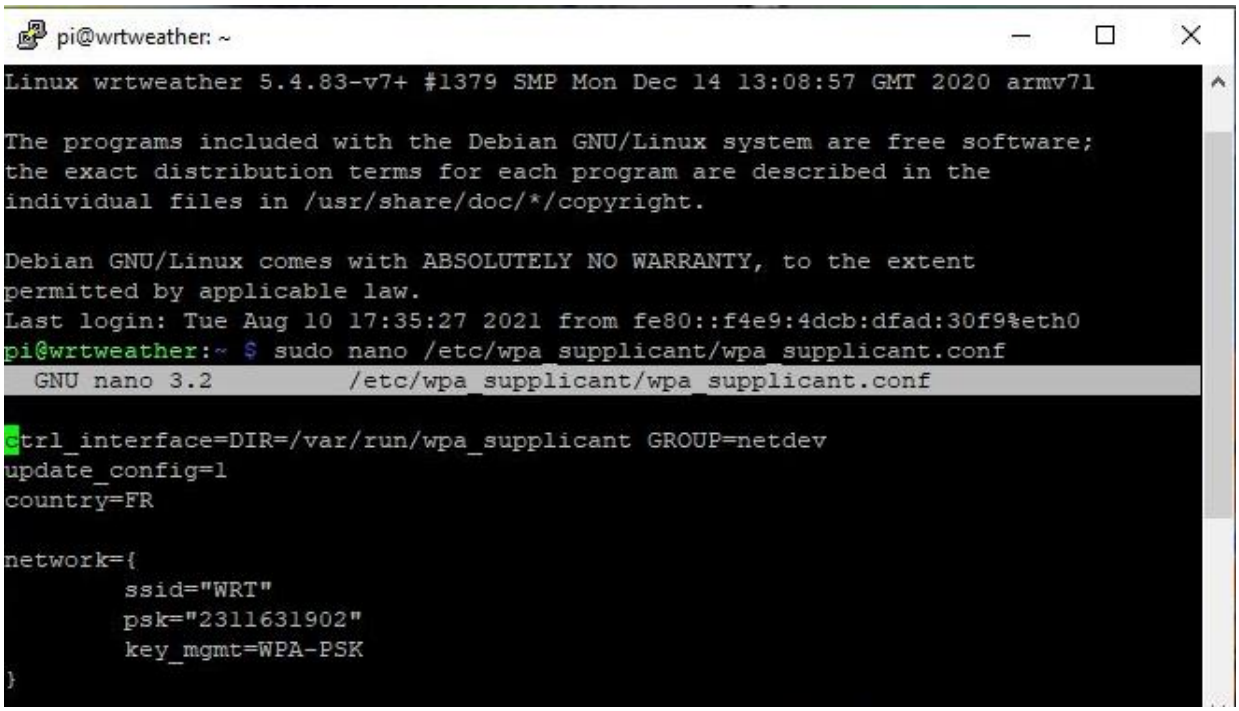
Press entre to modified the file. You must copy paste after the header:

Network= {

ssid = “Enter the name network here “

psk= “Enter the password “

}



For finish press “Ctrl O”, “Enter” and “Ctrl X”

Operating mode:

The Weather Station operate on the network for that you need connected your computer on same connexion. Install the divers sensors after to apply power. Push the reset button (3) for remove the old values (it is very useful before all the event). For shutdown, don’t forget the switch (4) before remove power.

Access on data:

There are three possibility:

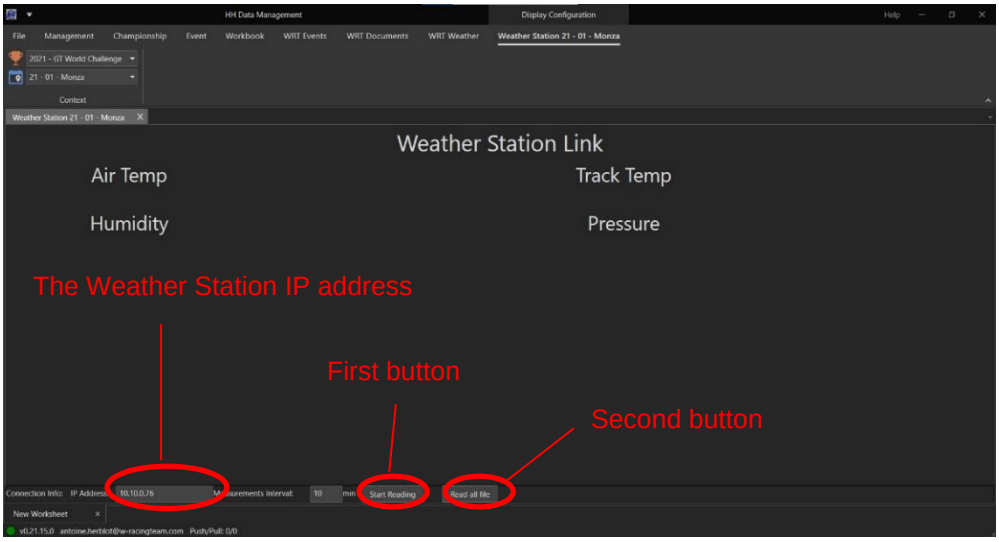
-HH Data management

-(IP address):9000

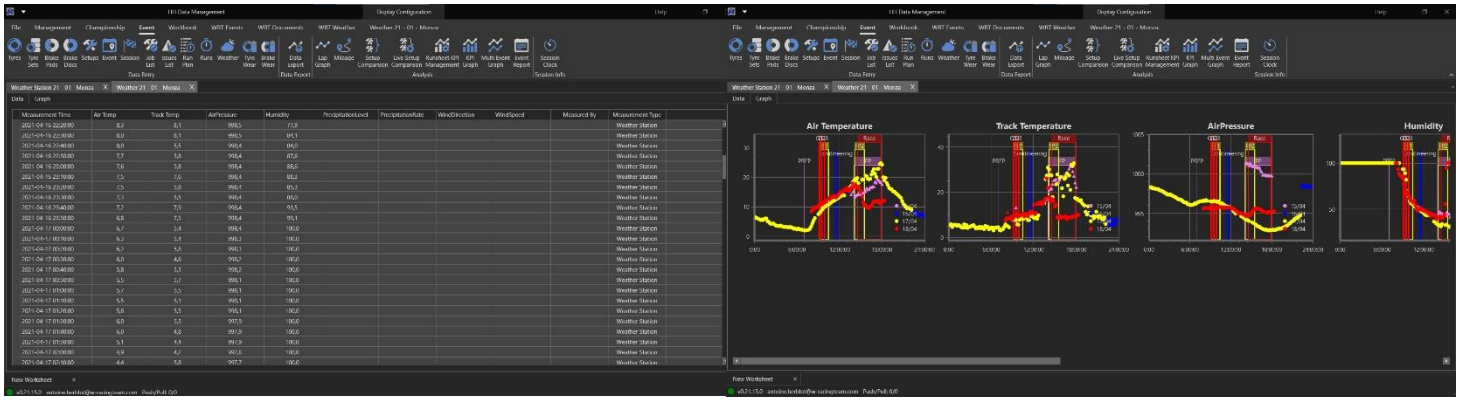
-(IP address):9001

For model : 10.10.0.76:9000
10.10.0.76:9001

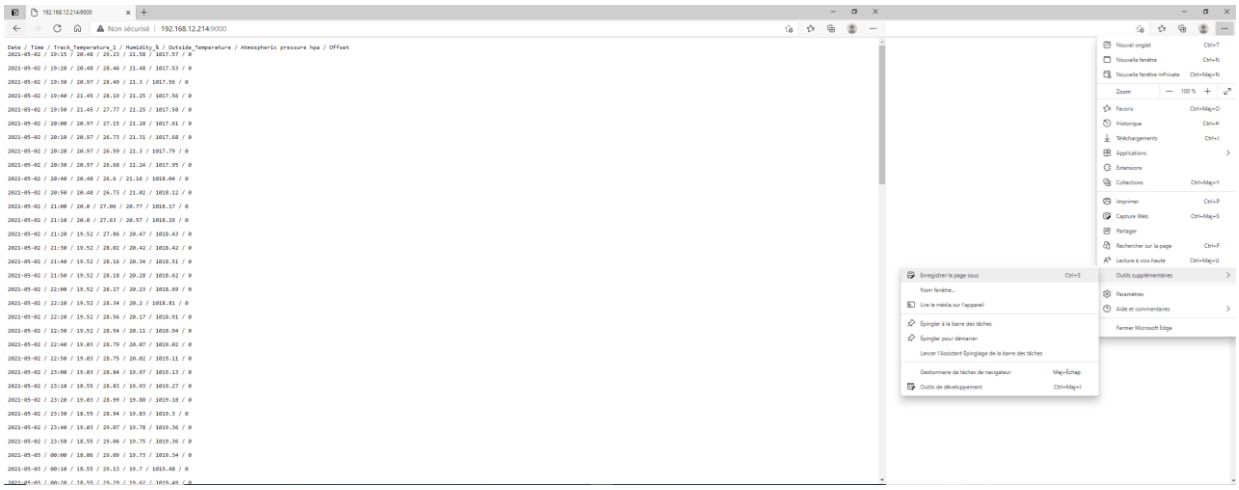
In HHDM open the Weather Station tab for informed the IP address at the bottom left. There are two buttons, the first allows you to start data acquisition at the interval of 10min. The second buttons take all values save on the Weather Station but **be careful because this operation overwrite your open database**.



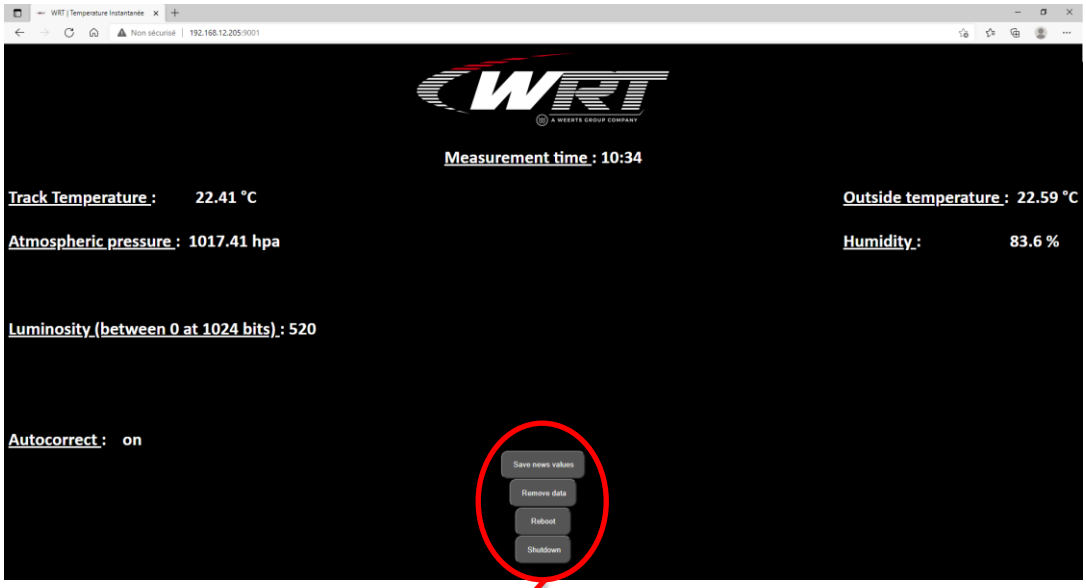
After the connection etablished at HHDM you have access to data on the event, weather tab.



If you open the web browser you can have access to other interfaces on the port :9000. On this, you have the data saved in the Weather Sation. You can also save as a .txt file.



On the port :9001, we have access to reel-time data when refresh the web page. Furthermore you find an input zone to apply a gain or an offset, if necessary.To finish we have a button for add a value when you want in the database.

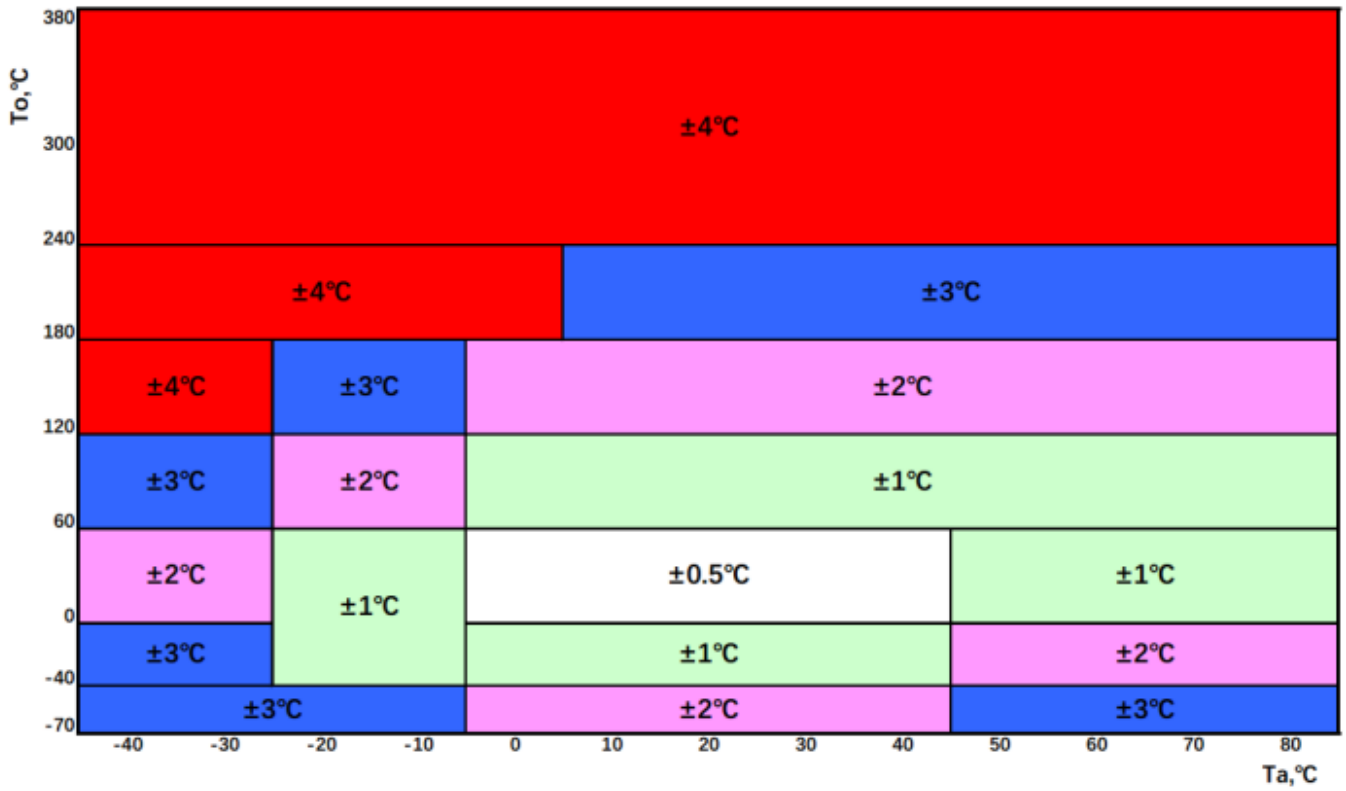


Botton for different action

LED statuts :

LED statuts	Signification
○ ○ ○	Green LED is off, system is off
○ ○ ☼	Green LED is flashing, the program reboot or take a news value. If the situation is persisting, you have a problem (restart system).
○ ○ ●	Green LED is on, the system function correctly.
○ ○ ○	Orange LED is off, all the sensors function correctly.
○ ☼ ○	
○ ● ○	Orange LED is on, you have over sensor problem.
○ ○ ○	Red LED is off, the Weather Station found a connection to transmit the data.
☼ ○ ○	Red LED is flashing, the program for remove the data in Weather Station is working
● ○ ○	Red LED is on, you have a connection problem.

Below is the accuracy of the sensor installed as a function of the measured temperatures and the outside temperature (tested and safe distance 1m60).



To is the measured temperature; Ta is the temperature of the environment the sensor locates in